

ShapTime: A General XAI Approach for Explainable Time Series Forecasting

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Abstract. The application of Explainable AI (XAI) in time series forecasting has gradually attracted attention, given the widespread implementation of machine learning and deep learning. ShapTime - A general XAI approach based on Shapley Value specially developed for explainable time series forecasting, which can explore more plentiful information in the temporal dimension, instead of only roughly applying traditional XAI approaches to time series forecasting as in previous works. Its novel components include: (1) It provides the relatively stable explanation in the temporal dimension, that is, the explanation result can reflect the importance of time itself, which is more suitable for time series forecasting than traditional XAI approaches; (2) It builds the practical application scenario of XAI - improving forecasting performance guided by explanation results. This is distinctly different from previous works, which only present the results of XAI as the demonstration of innovation. Eventually, in five real-world datasets, ShapTime's average performance improvements for Boosting, RNN-based and Bi-RNN-based reached 18, 20 and 35%, respectively.

Keywords: Time-series forecasting · Explainable AI · Shapley value

1 Introduction

Numerous time series forecasting competitions including M4 [1] and M5 [2] have shown that ML and DL perform significantly better than traditional statistical methods, especially for more complex tasks. This has led to research on the application of Explainable AI (XAI) in time series forecasting. Explainable time series forecasting aims to improve the trustworthiness of ML and DL in fields such as Finance, Energy and Meteorology. There are two main approaches to apply XAI in time series forecasting models: (1) directly using the existing model-agnostic method with high generality; (2) developing a model-specific method specifically for the model. These two approaches directly caused two key problems.

Problem 1. *In time series forecasting, the existing model-agnostic method is roughly applied, resulting in insufficient explanation.*

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The essential reason for the insufficient explanation is that most of the existing model-agnostic methods are feature attribution methods, given that XAI was originally developed based on regression and classification tasks, such as SHAP [3], LIME [4], etc. In contrast to time-series data, there is no temporal relationship among data instances for regression and classification, so the corresponding model-agnostic method pays more attention to feature importance (or contribution). However, this is not sufficient for time series forecasting. In time series forecasting models $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \omega_t$; $\omega_t = b + k_1 x_1 + k_2 x_2 + \cdots + k_n x_n$. y_{t-i} is the historical data of the target variable, and ω_t is composed of features x_i and intercept terms b. These traditional model-agnostic methods can only output feature importance k_i, but not the importance of time itself ϕ_i. Actually, due to human cognitive inertia, it is common to use feature importance as the explanation result even in model-specific methods.

Problem 2. *The specially developed model-specific method needs to be embedded in the model, resulting in low generality and high application cost.*

Several works have noticed Problem 1 developed some explainable time series forecasting models that can show periodicity as well as trend, which acts on the temporal dimension. However, these methods are still not able to output ϕ_i, and the degree of explanation is limited. In addition to the above problems, there is also a common problem in the field of XAI, which is listed as Problem 3 in this work.

Problem 3. *Lack of application scenarios.*

In works involving XAI, usually, explanation results are only presented as an innovation. In a large number of previous works, the application scenarios of XAI are only mentioned in the introduction part, including helping users trust the model and developers debugging the model, but these scenarios are not implemented.

Overall, on the one hand, the current model-agnostic method with high generality cannot fully explain the time series forecasting task, that is, it cannot realize the explanation of the temporal dimension. On the other hand, the model-specific method that can achieve temporal dimension explanation to a certain extent also has limitations, and its low generality also increases the cost of use. Therefore, a general XAI approach to explainable time series forecasting is needed, given the increasing importance of ML and DL in time series forecasting.

ShapTime[1] realizes the attribution of time by calculating the Shapley Value [5] on the temporal dimension, and finally outputs the importance of time itself ϕ_i. Therefore, ShapTime can realize the explanation in the temporal dimension, and it belongs to the model-agnostic method, which means that it can be deployed on any forecasting model at a lower cost. The foundation of ShapTime is the Shapley Value, which is the basis for numerous attribution methods including SHAP. Shapley Value comes from cooperative game theory, which studies

[1] See https://github.com/Zhangyuyi-0825/ShapTime.

how to reasonably distribute the benefits to the players in the alliance, and it has been proved to have some good properties. Therefore, in recent years, the development of XAI methods around Shapley Value is trying to become a stable path [6], and our ShapTime is exploring explainable time series forecasting as a branch on this path. Its contributions include:

- It realizes time attribution in the temporal dimension, that is, the importance of time itself ϕ_i can be obtained
- As a highly general model-agnostic method, it can be deployed in any forecasting model
- Its explanation results can be used as a guide to improve the forecasting performance of the model.

Subsequently, in Sect. 2, related work is summarized. In Sect. 3, the details of ShapTime are described. In Sect. 4, the experimental results are presented, including explanation results in the temporal dimension, forecasting performance improvements, and evaluation metrics. Section 5 is the conclusion.

2 Related Works

Previous approaches to explainable time series forecasting can be divided into three categories: Model-Agnostic methods, Model-Specific methods and Hybrid methods.

Model-Agnostic methods. It is realized by perturbing the input data set and inducing the change of the output, and finally attributing this change to the input features, so as to realize the explanation of the model. This is one approach that is widely used compared to others because of its generality(e.g. [7–11]). However, this type of XAI method was originally developed based on classification and regression tasks, so they often output feature importance instead of the importance of time itself, that is, they cannot output ϕ_i.

Model-Specific methods. This is a specific method for the development of time series forecasting models, that is, to embed the interpretation function into the model (e.g. [12,14]), in order to achieve better interpretation effect for time series forecasting. Although some works pay attention to the explanation of the temporal dimension (e.g. ([13,15]), they are still based on features, that is, the XAI approach outputs the feature importance at each moment and stitches them together to achieve the explanation of the temporal dimension. Strictly speaking, such an explanation also fails to output ϕ_i, and seriously lacks generality.

Hybrid methods. This is a approach to achieve explanation effects by hybridizing modules with a certain degree of interpretation function into time series forecasting model. The most representative approach is to hybridize the Attention Mechanism into the time series forecasting model (e.g. [16–18], and achieve the explanation through the explainability of the Attention Mechanism. Some works have discussed the explainability of the Attention Mechanism. Even though there are some controversies (e.g. [19]), researchers still hold a positive attitude towards its explainability (e.g. [20,21]. The Hybrid methods suffers from

the same problem as above approaches, that is, even if there is an explanation in the temporal dimension, this explanation is dependent on features. On the other hand, it also requires the development of new models and thus lacks generality.

3 Proposed Framework

3.1 Shapley Value for XAI

Shapley Value is one of the classic theories in cooperative games, which aims to distribute the benefits fairly to the players in the alliance. There is the correspondence between Shapley Value and model explanation: the features used for training correspond to "players", and the model's predictions correspond to "revenues". Therefore, the assignment achieved by Shapley Value can attribute the prediction result to the features, that is, the contribution (importance) of each feature to the prediction result. The assignment is achieved by the following formula:

$$k_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} \times (v(S \cup \{i\}) - v(S)). \tag{1}$$

where N is the set of all players (features) (1,2,3...i...n), which is the complete set; S is a subset of N, in which removes the explained feature i, with a total of 2^N; v is the gain function ($v(S) = E_{\hat{D}}[f(x) \mid x_S]$, where $\hat{D}$ is the empirical distribution of the training data and f is the black-model).

The reason why Shapley Value has become the basis of many XAI approaches is that it has some desirable properties, including: Efficiency, Symmetry, Linearity and Null player, and among many attribution methods, it is the only mapping ($v : 2^N \to \mathbb{R}$) that can satisfy the above properties.

Efficiency: The sum of all feature attributions is equal to the predicted result of the model (the total revenue of the alliance), which is a prerequisite for the establishment of the attribution method.

$$\sum_{i \in N} k_i = v(N) \tag{2}$$

Symmetry: If feature i and feature j can be substituted for each other, then their attribution results are also equal. This is the natural property that should be possessed as the attribution method.

$$v(S \cup \{i\}) = v(S \cup \{j\}) \tag{3}$$

Linearity: If there is the combination of two predictive models, then the contribution of the features within the combination should be equal to the sum of the contributions from the respective models. This property opens up possibilities for Shapley Value applications within hybrid frameworks.

$$k_i(v + w) = k_i(v) + k_i(w) \tag{4}$$

$$k_i(av) = ak_i(v) \tag{5}$$

Null player: If the value of a feature is meaningless (including null values or all equal values), then the feature will not have any impact on the attribution process.

$$v(S \cup \{i\}) = v(S) \tag{6}$$

Shapley Value has two key elements: "player" and "gain function". When both are determined, Shapley Value has the possibility to be calculated theoretically. Accordingly, Shapley Value is the attribution of "player".

3.2 ShapTime

In order to achieve attribution on the temporal dimension, time points should be considered as "players". Corresponding to the time series forecasting model, y_{t-i} is regarded as a "player", and after defining the gain function, the attribution of y_{t-i}, namely ϕ_i, can be realized through Shapley Value. The modified formula (ShapTime) is:

$$\begin{aligned} \phi_{t_i} = \sum_{S' \subseteq T \setminus t_i} & \frac{|S'|!(|T| - |S'| - 1)}{|T|!} \\ & \times (v'(S' \cup t_i) - v'(S')). \end{aligned} \tag{7}$$

where t_i is super-time (players), T is the set of all super-time. S' is a subset of T, in which removes the explained super-time t_i, with a total of 2^T. v' is the gain function of ShapTime.

Super-Time The so-called temporal attribution is to attribute the forecasting results to time, which means treating the time points as "players", which will inevitably cause the dimension explosion and the system crash. A similar problem is encountered in the XAI approach for image recognition—a large number of pixels makes the computational cost increase exponentially. In order to solve this problem, [4] proposed the concept of super-pixel, which is to cluster pixels with high similarity into a whole, and then participate in the calculation of pixel contribution, thereby greatly reducing the calculation cost.

In view of the similarity of the problem, we refer to his method to construct super-time in the temporal dimension, that is, the data set is divided into n super-times t_i according to the temporal dimension.

$$t_i = \{y_{t-i}, y_{t-i-1}, y_{t-i-2}, \ldots\} \tag{8}$$

In ShapTime, super-time is equivalent to "player". Crucially, although super-time controls the computational cost within an acceptable range, considering the complexity of $O(2^n)$ in the ShapTime, we recommend that n not exceed 10 or 11. Algorithm 1 shows the construction process of Super-time.

Algorithm 1 Super-Time

Input: Original data set: X
Parameter: The number of super-time: n
Output: All the super-time t_i

Let $L = \text{int}(\text{length}(X)/n)$.
Let start = length(X) - $L \times n$.
Let $X_{used} = X\,[start:,:]$.
for i in range (n) **do**
    $t_i = X\,[i \times L : (i+1) \times L]$.
end for
return all the t_i

Algorithm 2 Gain Function of ShapTime

Input: Explained model: f; Super-time: t_i
Output: All the gain value: v'

Let $S' \subseteq T \setminus t_i$.
Let $T = \{t_0, t_1, \cdots, t_n\}$
for S' in T **do**
    if length(S') == 1 **then**
        $S' = \{t_i\}$.
        $v' = \text{sum}(f(t_i))/\text{length}(t_i)$.
    else
        $S' = \{t_i, t_j, \cdots\}$.
        $S_c = concat(t_i, t_j, \cdots)$
        $v'(S_c) = \text{sum}(f(S_c))/\text{length}(S_c)$.
    end if
end for
return 2^T gain values: v'

Gain Function In ShapTime, the attributed object is super-time, which is the collection of time points within a period of time, that is, the "player" is no longer a time point at this time, but a time period. Correspondingly, our forecasting target y_t is also the collection of time points. However, the calculation of Shapley Value requires that for each "player" combination, the corresponding gain function outputs a value. Therefore, here, the averaged model f forecasting results are taken as the gain function v' of ShapTime (lines 6 and 10 in Algorithm 2). The formula is:

$$v'(S') = avg(f_{S'}(x_{S'})) \tag{9}$$

Eventually, the gain value v' of all combinations of super-time will be obtained, a total of $2^{|T|}$. In this way, according to Eq. (7), the forecasting result can be attributed to each super-time t_i, that is, ϕ_{t_i}, so as to realize the explanation in the temporal dimension.

Performance Improvement Even though research on explainability has received increasing attention in recent years, however, in general, explanation

results are simply presented as samples. This phenomenon does not only occur within the domain of explainable time series forecasting, but is the pervasive problem for the entire domain of Explainable AI (Problem 3). To explore the application scenarios of the XAI approach, we try to use ShapTime's explanation results as a guideline, aiming to improve the performance of time series forecasting.

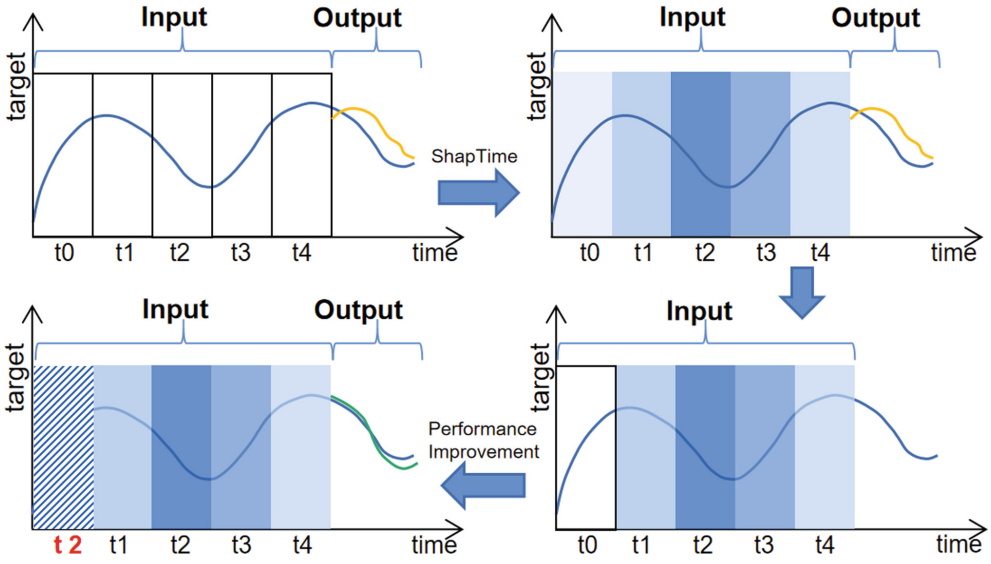


Fig. 1. ShapTime-based performance improvement process. The Blue line represents the original data set, and the orange line represents the forecasting results of the model. The blue background represents the explanation result of shaptime. The darker the color, the more important the t_i is, and vice versa. After obtaining the importance in the remporal dimension, the t_i with the lowest value is replaced by highest one. The green line represents the improved forecasting results

First of all, the input data set is divided equally into several super-times, and then, the importance of each super-time is calculated using ShapTime and visualized by the heat map, where the darker color means more important and the lighter color is less important. The way to improve performance is to replace low-importance t_i with high-importance t_i, and use this newly constructed data set to retrain the model. The reason for constructing such the improvement scheme (Fig. 1) is that ShapTime based on Shapley Value is the model-agnostic method, so its operation logic is based on the data set. In detail, the disturbance of the input data will change the forecasting result, and this type of method realizes attribution through this change. In fact, $(v'(S' \cup t_i) - v'(S'))$ represents this change. Therefore, dataset-specific refinements are naturally adopted when utilizing the explanation results output by this method.

Evaluation Criteria Currently, there is the lack of recognized correct labels in the XAI research field, which poses challenges for the evaluation of XAI methods. On the other hand, our ShapTime is the model-agnostic method for the explanation of the temporal dimension. In the field, this form of explanation is scarce, resulting in the lack of objects for comparison, so evaluation by contrast is difficult to achieve. Therefore, we propose some reasonable evaluation criteria to evaluate ShapTime effectively to a certain extent. It also provides a basic benchmark for subsequent research.

Property 1. *Under the premise of using the same data set, the explanation results of the XAI approach to similar types of models should be consistent to a certain extent.*

Property 2. *The XAI approach should be subject to sensitivity analysis, that is, when vital "players" are perturbed, there is the significant drop in forecasting performance.*

XAI's evaluation criteria are critical for users to build trust in machines. The most important of these is that the explanation results of XAI should maintain a certain degree of stability (Property 1), which is the basis for building trust. On this basis, the XAI method also guarantees validity (Property 2), that is, when important "players" are changed, the performance of the model will decline significantly. This can prove to a certain extent that the explanation results of this XAI approach are valid.

4 Experiments

In order to verify the practical application ability of ShapTime, we selected 5 real data sets, including: Climate Data, Energy Consumption, Solar Power, Gold Price, Tesla Stock. Among them, the first three are periodic data, and the last two are trend data, so as to test the performance of ShapTime under different data types.

On the other hand, there are two types of black box models involved in training: Boosting model and RNN-based model. The former consists of XGBoost and LightGBM; the latter includes RNN, LSTM, GRU (RNN-based) and Bi-RNN, Bi-LSTM, Bi-GRU(Bi-RNN-based). These forecasting models basically include the current mainstream methods in competition and practice. Their forecasting performance measures are shown in Table 1.

4.1 Explanation

We applied ShapTime on the above eight models and generated the explanation results (Fig. 2: For the sake of brevity, we only show the explanation results of ShapTime for XGBoost and LSTM in five data sets, and the complete explanation results are in the Github).

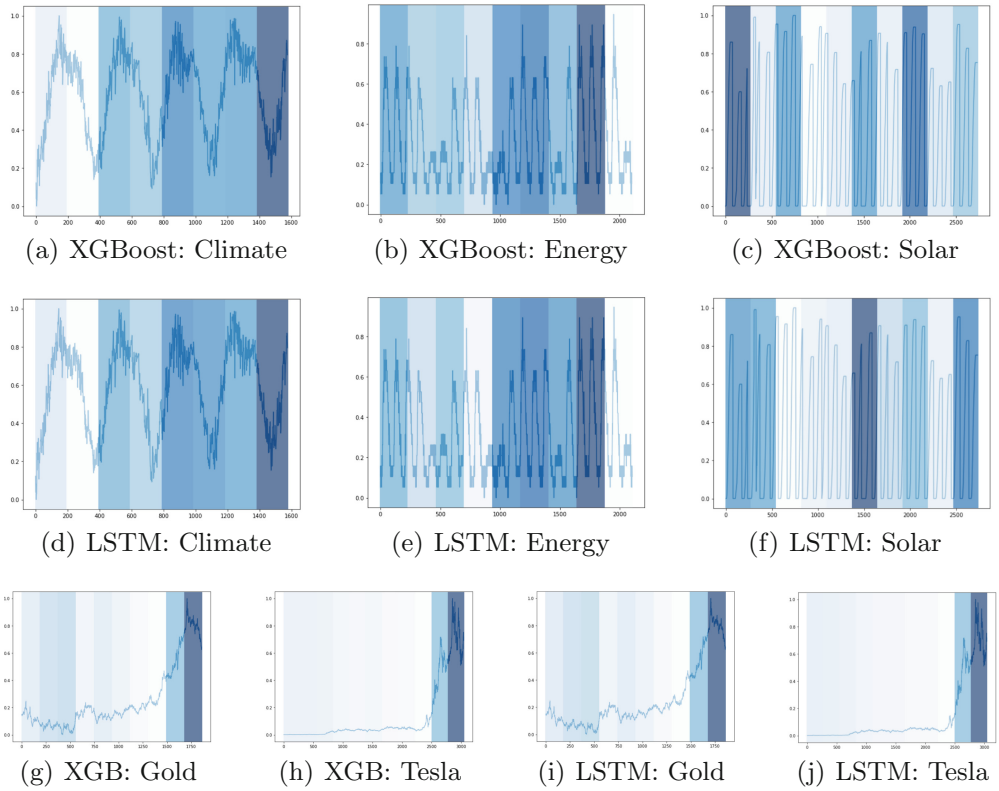
Table 1. Metrics to forecasting performance. forecasting models are divided into three categories: boosting; RNN-based; Bi-RNN-based. r^2 and MSE are used as forecasting performance metrics, for each model, the first row is r^2 and the second row is MSE. Within each class of models, the best model is marked (in Bold); among all models, the best model is additionally marked (in Brackets). As can be seen from the table, Bi-RNN-based performs best in trending data (Gold Price and Tesla Stock), and boosting performs best in periodic data (Daily Climate, Energy Consumption, and Solar Generation)

Data	Climate	Energy	Gold	Solar	Tesla
XGB	0.7756	0.7570	**0.7988**	0.9307	0.7632
	(0.0081)	0.0095	0.0006	0.0055	0.0029
LGB	**(0.7814)**	**(0.8290)**	0.7779	**0.9322**	**0.7777**
	0.0082	**(0.0071)**	**0.0006**	**0.0053**	**0.0023**
RNN	0.7170	0.6574	0.7018	0.9587	0.8133
	0.0104	0.0128	0.0011	0.0051	0.0022
LSTM	**0.7507**	0.6249	**0.8182**	0.9544	**0.8586**
	0.0081	0.0153	**0.0005**	0.0057	**0.0017**
GRU	0.6719	**0.7130**	0.7096	**(0.9661)**	0.8176
	0.0117	**0.0119**	0.0009	**(0.0035)**	0.0025
Bi-R	**0.7257**	0.6804	0.8048	0.9312	0.7990
	0.0102	0.0120	0.0008	0.0099	0.0029
Bi-L	0.6903	**0.7122**	**(0.8791)**	0.9326	**(0.8689)**
	0.0103	**0.0119**	**(0.0005)**	0.0082	**(0.0016)**
Bi-G	0.7136	0.5927	0.8664	**0.9481**	0.7055
	0.0109	0.0154	0.0006	**0.0056**	0.0045

In explanation results, the Daily Climate dataset (Fig. 2a–f) is taken as the example. In this example, the number n of super-time is set to 8, then the corresponding theoretical modified time series model is:

$$y_t = \phi_{t_0} t_0 + \phi_{t_1} t_1 + \cdots + \phi_{t_7} t_7 + \omega_t \tag{10}$$

After the training is completed, we use ShapTime to explain XGBoost and LSTM, so as to attribute the forecasting results of the model to each super-time, and then get each ϕ_{t_i} value, and display them visually through the heat map. We can clearly see that both models capture the most recent super-time t_7 as the most important input. In the Energy Consumption dataset, the two models regard t_7 as the most important super-time; in Gold Price it is t_9; in Tesla Stock it is t_{10}. Different situations appear in Solar Generation. The super-time captured by the two models is different. XGBoost captures t_0 as the most important, while LSTM captures t_5.



(a) XGBoost: Climate (b) XGBoost: Energy (c) XGBoost: Solar

(d) LSTM: Climate (e) LSTM: Energy (f) LSTM: Solar

(g) XGB: Gold (h) XGB: Tesla (i) LSTM: Gold (j) LSTM: Tesla

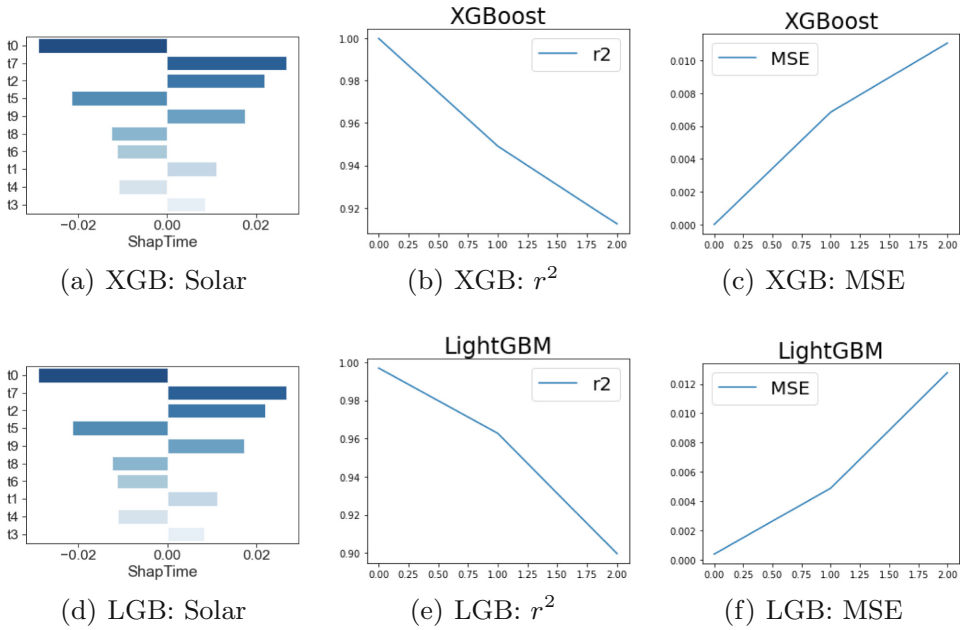
Fig. 2. Example of explanation results from shaptime. This is shaptime's explanation of XGBoost and LSTM. The heat map is used to visualize the explanation results of the input data. The darker the color, the more important the super-time. This means that the model pays more attention to this super-time during the training process. It can be seen from figures that the explanation of trend data by LSTM and XGBoost is basically consistent, and there is a certain degree of difference in the explanation of periodic data, especially in the solar generation data set

Looking at all the 5×8 explanation results, the most important super-time captured by all models is the last one in the trend data, but in the periodic data, there is no obvious rule. However, if we analyze the forecasting performance of the model (Table 1), we can still find some potential rules, that is, the Boosting model is more suitable for periodic data, while the RNN-based model is more suitable for periodic data when the explanation results are generally consistent.

4.2 Evaluation

According to the above evaluation criteria, we evaluate the explanation results of ShapTime, and the evaluation examples are shown in Fig. 3. Similarly, for simplicity, we only display the evaluation results on the Boosting model. Figure 3a–d represent the explanations of ShapTime for XGBoost and LightGBM, respectively. It can be intuitively seen that the explanation of ShapTime can be maintained all the time for the same type of models, which is a common pattern

in all the explanation results. This is key to the trust of users from all industries. Imagine if the explanation results change frequently during use, which will directly lead to user distrust or even abandonment.



(a) XGB: Solar (b) XGB: r^2 (c) XGB: MSE

(d) LGB: Solar (e) LGB: r^2 (f) LGB: MSE

Fig. 3. Evaluation Example of ShapTime Explanation Results. **a**, **b** Satisfy Property 1, **b**, **c**, **e**, **f** Satisfy Property 2

This is exactly the purpose for which Property 1 was defined. However, in case of different types of models, in principle, differences in the explanation results need to be allowed, since models of varying architectures are not equally sensitive to the nature of the distribution of the data. As in Fig. 2c–h, the explanation results present differences in the various models.

On the other hand, theoretically, if we perturb the training data according to the explanation results, i.e., the important super-time is replaced with the least important, then the forecasting performance of the model will be significantly decreased. In this example, the most important t_0 is replaced with the least contributing t_3, as well as t_7 is replaced with t_4. Correspondingly, the r^2 and MSE of XGBoost and LightGBM exhibit significant and gradual performance degradation (Fig. 3b, c, e, f). This evaluation schema is the relatively classical evaluation method in the field of XAI, and it is also known as sensitivity analysis. In this work, it is summarized in Property 2.

4.3 Improvement

To create valuable XAI application scenarios, we use the explanation results of ShapTime as our guide for achieving improved forecasting performance, and the

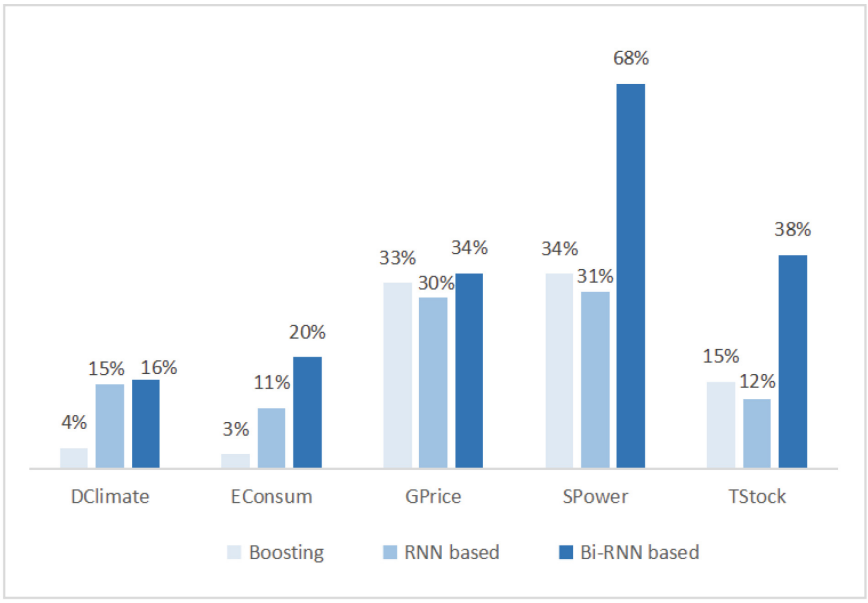
improvement process is shown in Fig. 1. Table 2 shows the improved performance metrics (compared to Table 1), and the results show that Bi-RNN-based and Boosting still maintain their original advantages in trending and periodic data, respectively.

Table 2. ShapTime guided forecasting performance improvement. r^2 and MSE are used as forecasting performance metrics, for each model, the first row is r^2 and the second row is MSE

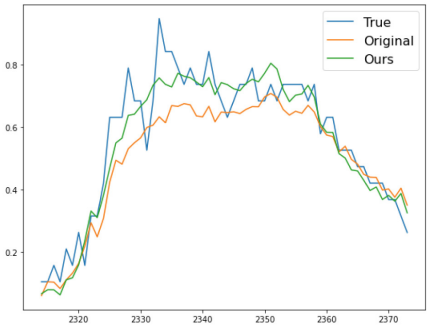
Data	Climate	Energy	Gold	Solar	Tesla
XGB	**(0.7898)**	0.7975	**0.8880**	0.9521	**0.8198**
	0.0075	0.0093	**(0.0004)**	0.0038	**0.0021**
LGB	0.7847	**(0.8445)**	0.8715	**0.9573**	0.8038
	0.0082	**(0.0069)**	0.0004	**0.0033**	0.0022
RNN	0.7255	**0.7572**	**0.8535**	0.9701	0.8385
	0.0096	0.0116	0.0006	**0.0024**	0.0020
LSTM	**0.7751**	0.6412	0.8287	0.9597	**0.8812**
	(0.0070)	0.0142	**0.0005**	0.0047	**0.0016**
GRU	0.7470	0.7376	0.7825	**0.9740**	0.8498
	0.0092	**0.0101**	0.0006	0.0026	0.0020
Bi-R	0.7428	0.7425	0.8980	0.9693	0.8932
	0.0082	0.0101	0.0004	0.0036	0.0016
Bi-L	**0.7532**	**0.7604**	0.9002	0.9742	**(0.9080)**
	0.0084	**0.0100**	0.0004	0.0028	**(0.0013)**
Bi-G	0.7412	0.7092	**(0.9043)**	**(0.9840)**	0.8505
	0.0100	0.0112	**0.0004**	**(0.0014)**	0.0021

The measure of the degree of improvement is presented in Fig. 4a, which shows the average degree of improvement for each type of model in each data set. The percentages are calculated based on the degree of improvement in Table 2 compared to Table 1. It can be intuitively seen that ShapTime shows the maximum improvement for Bi-RNN-based, and the most significant improvement for Solar Generation. Overall, the effect of ShapTime on Boosting is less significant than that of other types of models. Specifically, ShapTime has the most significant improvement on Bi-GRU. In the original forecasting performance (Table 1), Bi-GRU does not possess the best performance in all datasets, while after the improvement (Table 2), the best forecasting performance of Gold Price and Solar Generation appears in Bi-GRU.

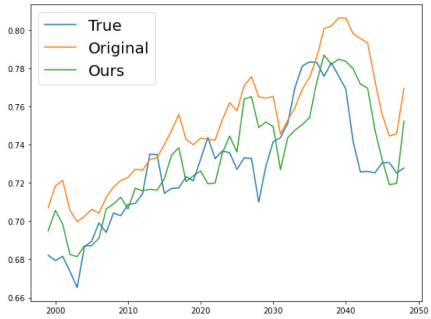
The partial visualization of the improvement effect is shown in Fig. 4b–e. From the forecasting effect, the forecasting results after ShapTime improvement still maintain approximately the same pattern as the original forecasting results, however, they can be much closer to the original data, thus achieving the performance improvement.



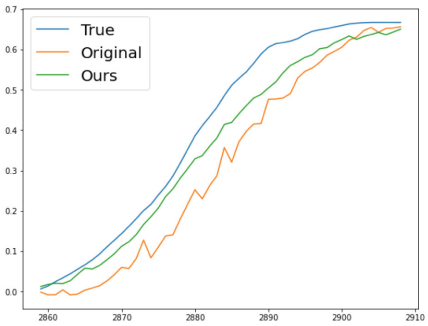
(a) Improvement results of forecasting performance



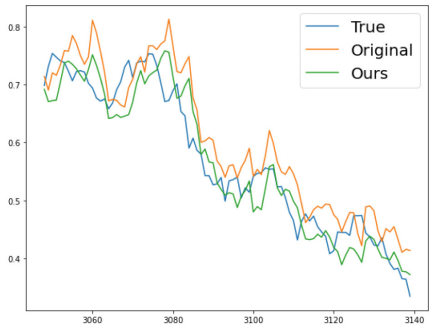
(b) Improve Energy



(c) Improve Gold



(d) Improve Solar



(e) Improve Tesla

Fig. 4. Visualization of performance improvement. "Original" is the output of the original forecasting model, and "Ours" is the output of the shaptime-guided forecasting model

5 Conclusion

In this work, the XAI approach specifically oriented to time series forecasting is developed, and we name it ShapTime since its computation is based on Shapley Value. It enables attribution in the temporal dimension, thus explaining the importance of time itself, which differs from previous works.

With the explanation of ShapTime, we are able to understand the forecasting model to some extent. In trending time series data, all models focus on the most recent data as the most important learning object, while in periodic time series data, this pattern does not obviously exist and different models do not necessarily focus on the same time period for learning.

On the other hand, with the help of ShapTime explanation, we have been able to achieve the improved performance in time series forecasting. By replacing data in times of low contribution with high ones, performance improvements can be achieved to some extent. The improved performance metrics show that the Boosting model and the Bi-RNN-based model are still able to maintain their original advantages in periodic data and in trending data, respectively. In summary, ShapTime showed the most significant improvement for the Bi-RNN-based model, with the average improvement of 35%. In particular, ShapTime showed the most significant improvement for the Bi-GRU model, with the 73.87% improvement in the Solar Generation dataset.

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